AI101: Data Science and AI Essentials with Python | School of AI and Data Engineering, IIT Ropar

SMS Spam Detection

using Logistic Regression

Mini-Project (team of 4) | Team no.: 48

Member	Entry no.	GitHub handle	Role
Sadhil Khandelwal	2026EEB1404	2026eeb1404-stack	Documentation Lead
Abhinav Singh	2026EEB1315	Obliterator-Abhinav	Execution Lead
Aadrish Deb	2026MCB1503	2026mcb1503-pixel	PPT Lead
Kusulure Sri Charan	2026AIB1052	K-SriCharan-7781	Video Lead

Repo: https://github.com/Obliterator-Abhinav/sms-spam-detection

Video: https://www.loom.com/share/b74e7307e0224af4ac9aa607958498e8

Problem and Data

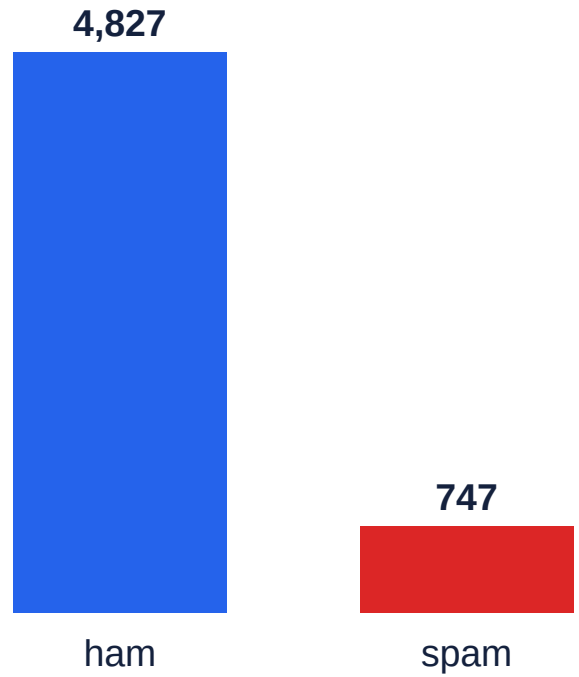
The problem

- Decide if an SMS is spam (unwanted ad or scam) or ham (a normal message).
- A filter must block scams without blocking real messages. A wrongly blocked real message is usually worse than one spam getting through.
- Binary classification: spam = 1 (positive class), ham = 0. Model: Logistic Regression.

The data

- SMS Spam Collection, UCI ML Repository (CC BY 4.0). Almeida & Hidalgo (2011).
- 5,574 messages, 2 columns: label and message. No missing values.
- 403 duplicates removed, leaving 5,171 (4,518 ham, 653 spam).
- Extra numeric features: message length, digit count, capital-letter ratio, exclamation marks.

Class balance (original data)

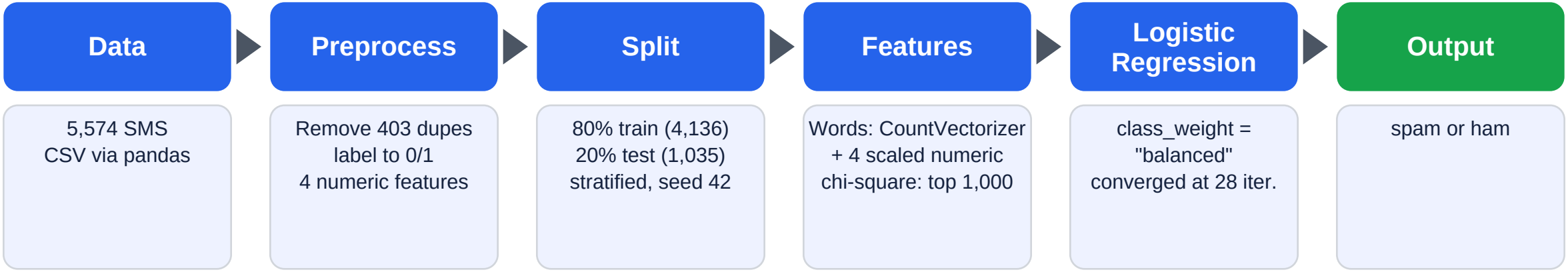


About 13% spam. Always saying "ham" gives ~87% accuracy, so accuracy alone is misleading.

Approach

Pipeline

Scaler and vectoriser are fitted on training data only.

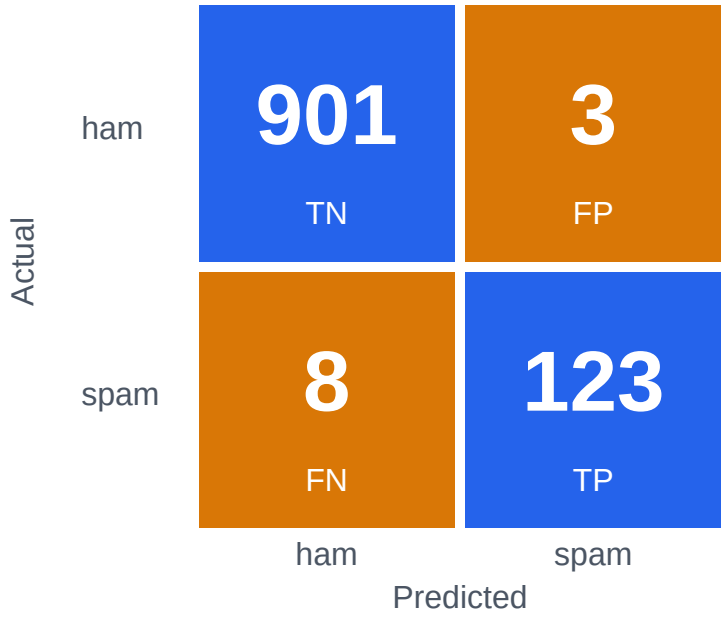


Final input and libraries

- 1,004 columns per message: 1,000 words + 4 numeric features (3,310 words before selection).
- Python 3.12 / 3.13, pandas, numpy, scikit-learn, scipy, matplotlib, seaborn, Jupyter Notebook.

Results

Confusion matrix (test set, 1,035 messages)



0.9894	0.9762	0.9389	0.9572
Accuracy	Precision (spam)	Recall (spam)	F1-score (spam)

- Only 3 real messages wrongly sent to spam (false positives), which matters most for a spam filter.
- Found 123 of 131 spam messages; missed 8.
- Train F1 0.983 vs test F1 0.957: small overfitting, test results still strong.
- Balanced class weights raised recall from 0.908 to 0.939.

Output of the evaluation cell after Run All Cells

```
Results on the training data
Accuracy : 0.9956
Precision: 0.9755
Recall   : 0.9904
F1-score : 0.9829

Results on the testing data
Accuracy : 0.9894
Precision: 0.9762
Recall   : 0.9389
F1-score : 0.9572
```

Reproduce: pip install -r requirements.txt, then run SMS_Spam_Logistic_Regression.ipynb (Run All Cells).

Learnings

What worked

- Word counts + 4 numeric features.
- digit_count was the strongest signal (coef. 3.10; spam averages ~15 digits, ham ~0.3).
- Balanced class weights.
- High precision: only 3 false positives.

What is limited

- Missed 8 spam: short or normal-looking texts, one with scrambled spelling.
- Dataset quirks: "lt"/"gt" come from <#> in 197 ham messages; "arsenal" learned from 2 spam messages.
- Data is from ~2011 and in English; today's Indian spam (OTP, Hinglish) not tested.
- Settings not tuned; word counts ignore order.

Who did what

- Sadhil Khandelwal: Documentation Lead - README and project docs.
- Abhinav Singh: Execution Lead - data pipeline in the notebook, requirements.txt and fresh-clone testing.
- Aadrish Deb: PPT Lead - the 5 slides.
- Kusulure Sri Charan: Video Lead - the project video and code demo.
- All four members contributed code (see commit history).